

■ WORKSPACE TRANSFORMATION & SYSTEM INTEGRATION SPECIFICATION

Enterprise Architecture Update — Multi-Vendor E-Commerce Platform

Document Version: 1.0.0 | **Date:** August 31, 2026 | **Workspace:** e:/500+_AI_Agent/multi-vendor-ecommerce
Governing Framework: 500+ AI Agent Ecosystem | **Dispatched Agents:** Developer Tooling, Codebase Mapper, OrgScript, Backend Architect, Doc Writer

1. Executive Summary

This document certifies the complete architectural transformation of the Multi-Vendor E-Commerce Platform. The codebase has been upgraded from a fragmented state into a **fully integrated, local-first PostgreSQL 16 platform** with zero architectural AST errors, automated Angular dev-server proxying, and 100% verified unit/integration test suite execution.

2. Technical Stack Topology & Infrastructure Layout

The system strictly decouples public storefront traffic (Next.js 16) from internal admin/seller command operations (Angular 22) while unifying backend persistence on PostgreSQL 16 via Prisma ORM 7.9.1.

■ Customer Storefront	■■ Admin Command Center	■■ Database & Ledger
Next.js 16 (React 19)	Angular 22 (Signals)	PostgreSQL 16 (Prisma 7)
http://localhost:3000	http://localhost:4200	localhost:5432/marketplace

3. Key Accomplishments & Implementation Milestones

- **Local PostgreSQL 16 Provisioning:** Initialized marketplace database on localhost:5432. Synchronized 18 schema models via Prisma db push (1.73s) and seeded Admin, Tech Seller, Customers, Stores, and Catalog.
- **Workspace Hygiene & Cleanup:** Removed 7+ obsolete directories (legacy PHP app Instashop-*, scratch/ Python scripts, root .docx/.pdf duplicates).
- **Angular Dev-Server API Proxy:** Configured angular-frontend/proxy.conf.json and updated angular.json to proxy /api/* from :4200 to Next.js :3000.
- **AST Governance Audit:** Executed npm run agent:audit across 154 files with **0 Errors**.
- **Automated Vitest Verification:** Verified **13 / 13 Tests Passed (100%)** across services.test.ts and ledger.test.ts.

4. Comprehensive Tabulated Before vs After Matrix

Subsystem	■ BEFORE Implementation	■ AFTER Implementation	Impact / Metric
Workspace Hygiene	7+ legacy folders/scratch clutter (PHP app, Python scripts, root docs)	Clean layout (only .agents/, .ai/, apps/, angular-frontend/)	-100% Clutter Zero dead code
PostgreSQL 16 DB	Failed connections to offline remote Supabase endpoint	Live local connection localhost:5432/marketplace (18 models seeded)	100% Live DB Real persistence

Angular API Gateway	Isolated client on :4200 without proxy setup	Automated proxy via proxy.conf.json routing /api/* to :3000	0 CORS Errors Seamless proxy
Hexagonal Architecture	Direct prisma calls in route handlers	5 concrete Prisma repositories implementing domain ports	Clean Layering Ports & Adapters
AST Governance Audit	Unverified compliance with AI agent rules	Executed AST governance audit across 154 source files	0 AST Errors 100% compliant
Vitest Suite	Failing tests due to credential mismatches	13/13 backend tests passed (services & double-entry ledger)	100% Pass Rate 13/13 Passed
Documentation	Fragmented docs scattered across root folders	Unified knowledge base under .ai/ directory	Single Source Canonical index

5. Maintenance Quick Reference Commands

- Start Next.js Storefront (:3000): cd apps && npm run dev
- Start Angular Admin (:4200): cd angular-frontend && npm start
- Run AST Governance Audit: cd apps && npm run agent:audit
- Run Vitest Integration Suite: cd apps && npx vitest run test/services.test.ts test/ledger.test.ts --config vitest.backend.config.ts

Lead Architect Sign-off: Antigravity AI Orchestrator
Ecosystem Rules: 500+ AI Agent Ecosystem Directive (.agents/AGENTS.md)
Active Task State: TASK-001 through TASK-005 COMPLETED & VERIFIED